



Machine Learning in Production

Measuring Fairness

Diving into Fairness...

Fundamentals of Engineering AI-Enabled Systems

Holistic system view: AI and non-AI components, pipelines, stakeholders, environment interactions, feedback loops

Requirements:

System and model goals
User requirements
Environment assumptions
Quality beyond accuracy
Measurement
Risk analysis
Planning for mistakes

Architecture + design:

Modeling tradeoffs
Deployment architecture
Data science pipelines
Telemetry, monitoring
Anticipating evolution
Big data processing
Human-AI design

Quality assurance:

Model testing
Data quality
QA automation
Testing in production
Infrastructure quality
Debugging

Operations:

Continuous deployment
Contin. experimentation
Configuration mgmt.
Monitoring
Versioning
Big data
DevOps, MLOps

Teams and process: Data science vs software eng. workflows, interdisciplinary teams, collaboration points, technical debt

Responsible AI Engineering

Provenance,
versioning,
reproducibility

Safety

Security and
privacy

Fairness

Interpretability
and explainability

Transparency
and trust

Ethics, governance, regulation, compliance, organizational culture

Reading

Required:

- Nina Grgic-Hlaca, Elissa M. Redmiles, Krishna P. Gummadi, and Adrian Weller. [Human Perceptions of Fairness in Algorithmic Decision Making: A Case Study of Criminal Risk Prediction](#) In WWW, 2018.

Quick schedule note

No lab this week: Carnival

No class next WEDNESDAY (Apr 15)

Learning Goals

- Understand different definitions of fairness
- Discuss methods for measuring fairness
- Outline interventions to improve fairness at the model level
- Reason about which definition is appropriate for a given context
- Recognize the limits of classifier-based measures for foundation models

Recall Breakout: College Admission

Scenario: Evaluate applications & identify students likely to succeed

Application materials: transcript, GPA, test scores (GRE/SAT), personal essays, letters of recommendation, resume/activities list, demographic information (gender, race, home address)

As a group, post to #1ecture, tagging members:

- **Possible harms:** Allocation of resources? Quality of service? Stereotyping? Denigration? Over-/Under-representation?
- **Sources of bias:** Historic bias?
- What to do about it, if any: ...

Recall: What is fairness?

Fairness discourse asks questions about how to treat people and whether treating different groups of people differently is ethical. If two groups of people are systematically treated differently, this is often considered unfair.

Fairness: Measurements

(Repeated) Caveat on Intersectionality

Individuals can and do fall into multiple groups!

Subgroup fairness gets *extremely* technically complicated quickly.

We therefore focus on the simple cases for the purposes of the material in this class.

Measuring fairness: then and now

Fairness research developed formal metrics for *classifiers*. These operate at the decision boundary of the model.

- Still relevant and widely used; we will start with them.

Foundation models are often used as classifiers in pipelines: we will discuss how the classic metrics apply to them (or not).

Foundation models *also* generate content. We will discuss what “fair” means there, and ideas about measuring it, after the classic metrics.

Classic Fairness Measurements

- Anti-classification (fairness through blindness)
- Group fairness (independence)
- Equalized odds (separation)

... and numerous others and variations!

Anti-Classification

Anti-Classification



- Also called *fairness through blindness* or *fairness through unawareness*
- Ignore certain sensitive attributes when making a decision
- Example: Remove gender and race from mortgage model

Anti-Classification: Example

Remote Appraisals of Homes Could Reduce Racial Bias

Desktop appraisals, in which an appraiser never meets a homeowner, could reduce discriminatory practices, such as undervaluing homes owned by Black people.

"After Ms. Horton removed all signs of Blackness, a second appraisal valued a Jacksonville home owned by her and her husband, Alex Horton, at 40 percent higher."

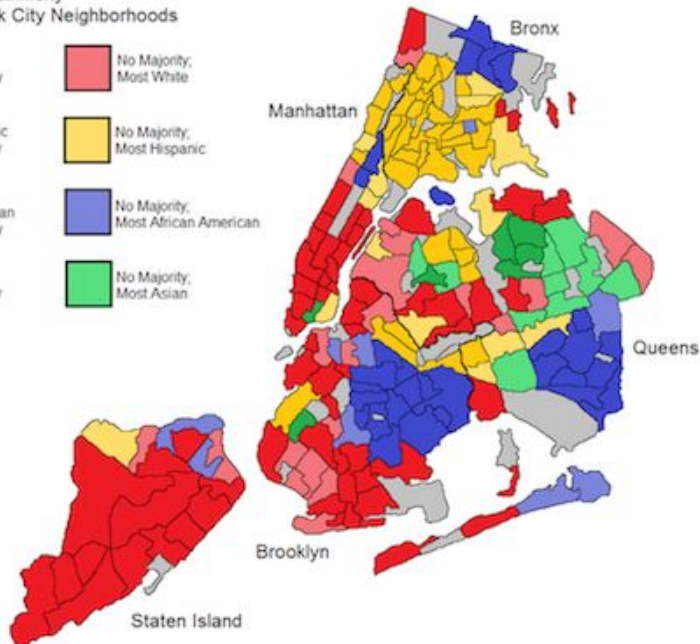
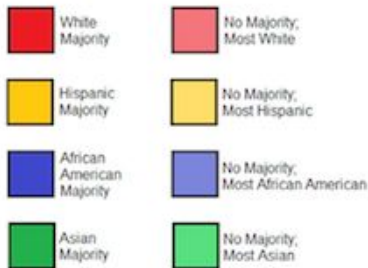
Anti-Classification



Easy to implement, but any limitations?

Limitation: Proxies

Race and Ethnicity
in New York City Neighborhoods



Features correlate with protected attributes

This is also true in foundation models; protected attributes are encoded across the embedding space.

Anti-Classification



- Ignore certain sensitive attributes when making a decision
- Advantage: Easy to implement and test
- Limitations
 - Sensitive attributes may be correlated with other features
 - Some ML tasks need sensitive attributes (e.g., medical diagnosis)

Ensuring Anti-Classification



How to train models that are fair w.r.t. anti-classification?

Ensuring Anti-Classification

How to train models that are fair w.r.t. anti-classification?

- Simply remove features for protected attributes from training and inference data
- Null/randomize protected attribute during inference
- Yes, you can also do that for foundation model-based classifiers

(does not account for correlated attributes, is not required to)

Testing Anti-Classification



How do we test that a classifier achieves anti-classification?

Testing Anti-Classification

Straightforward invariant for classifier f and protected attribute p :

$$\forall x. f(x[p \leftarrow 0]) = f(x[p \leftarrow 1])$$

(does not account for correlated attributes, is not required to)

- Test with *any* test data, e.g., purely random data or existing test data
- Any single inconsistency shows that the protected attribute was used. Can also report percentage of inconsistencies.

Breakout: Cancer Prognosis

	Male Patient Results			Female Patient Results	
	Actual cancer	Actually no cancer		Actual cancer	Actually no cancer
Predicted cancer	23	11	Predicted cancer	13	5
Predicted no cancer	41	925	Predicted no cancer	2	480

Individually, post to #1ecture:

Does the model meet anti-classification fairness w.r.t. gender?

$$\forall x. f(x[p \leftarrow 0]) = f(x[p \leftarrow 1])$$

Write your calculation and reasoning!

Anti-Classification: Discussion

Testing of anti-classification barely needed, because easy to ensure by constructing during training or inference!

- Anti-classification is a good starting point to think about protected attributes
- Useful baseline for comparison
- Easy to implement, but only effective if (1) no proxies among features and (2) protected attributes add no predictive power

Group fairness

Fairness: Measurements

- Anti-classification (fairness through blindness)
- *Group fairness (independence)*
- Equalized odds (separation)
- ...and numerous others and variations!

Group fairness

Key idea: Outcomes matter, not accuracy!

Compare outcomes across two groups

- Similar rates of accepted loans across racial/gender groups?
- Similar chance of being hired/promoted between gender groups?
- Similar rates of (predicted) recidivism across racial groups?

Disparate impact vs. disparate treatment

Disparate treatment: Practices or rules that treat a certain protected group(s) differently from others

- e.g., Apply different mortgage rules for people from different backgrounds

Disparate impact: Neutral rules, but outcome is worse for one or more protected groups

- Same rules are applied, but certain groups have a harder time obtaining mortgage in a particular neighborhood

Group unfairness in discrimination law

Relates to *disparate impact* and the four-fifths rule

Can sue organizations for discrimination if they

- mostly reject job applications from one minority group (identified by protected classes) and hire mostly from another
- reject most loans from one minority group and more frequently accept applicants from another

Four-fifths rule: If the selection rate for a protected group is less than 80% of the selection rate for the group with the highest selection rate, there is adverse impact.

Notation

- X : Feature set (e.g., age, race, education, region, income, etc.)
- $A \in X$: Sensitive attribute (e.g., gender)
- R : Regression score (e.g., predicted likelihood of on-time loan payment)
- Y' : Classifier output
 - $Y' = 1$ if and only if $R > T$ for some threshold T
 - e.g., Grant the loan ($Y' = 1$) if the likelihood of paying back $> 80\%$
- Y : Target variable being predicted ($Y = 1$ if the person actually pays back on time)

Setting classification thresholds: Loan lending example

Group Fairness

$$P[Y' = 1 | A = a] = P[Y' = 1 | A = b]$$

- Also called *independence* or *demographic parity*
- Mathematically, $Y' \perp A$
 - Prediction (Y') must be independent of the sensitive attribute (A)
- Examples:
 - The predicted rate of recidivism is the same across all races
 - Both women and men have the equal probability of being promoted
 - i.e., $P[\text{promote} = 1 \mid \text{gender} = \text{M}] = P[\text{promote} = 1 \mid \text{gender} = \text{F}]$

Group Fairness Limitations



Group Fairness Limitations

- Ignores possible correlation between Y and A
 - Rules out perfect predictor $Y' = Y$ when Y & A are correlated!
- Permits abuse and laziness: Can be satisfied by randomly assigning a positive outcome ($Y' = 1$) to protected groups
 - e.g., Randomly promote people (regardless of their job performance) to match the rate across all groups

Adjusting Thresholds for Group Fairness

Select different classification thresholds (t_0, t_1) for different groups ($A = 0, A = 1$) to achieve group fairness, such that

$$P[R > t_0 | A = 0] = P[R > t_1 | A = 1]$$

Example: Mortgage application

- *R: Likelihood of paying back the loan on time*
- *Suppose: With a uniform threshold used (i.e., $R = 80\%$), group fairness is not achieved*
 - $P[R > 0.8 | A = 0] = 0.4, P[R > 0.8 | A = 1] = 0.7$
- *Adjust thresholds to achieve group fairness*
 - $P[R > 0.6 | A = 0] = P[R > 0.8 | A = 1]$

Adjusting Thresholds for Group Fairness

Mortgage application: $P[R > 0.6 \mid A = 0] = P[R > 0.8 \mid A = 1]$

Wouldn't group $A = 1$ argue it's unfair? When does this type of adjustment make sense?



Testing Group Fairness

Collect realistic, representative data (not randomly generated!)

- Use existing validation/test data
- Monitor production data
- (Somehow) generate realistic test data, e.g. from probability distribution of population

Separately measure the rate of positive predictions

- e.g., $P[\text{promoted} = 1 \mid \text{gender} = M]$, $P[\text{promoted} = 1 \mid \text{gender} = F] = ?$

Report issue if the rates differ beyond some threshold ϵ across groups

Breakout Cont': Cancer Prognosis

Male Patient Results			Female Patient Results		
	Actual cancer	Actually no cancer		Actual cancer	Actually no cancer
Predicted cancer	23	11	Predicted cancer	13	5
Predicted no cancer	41	925	Predicted no cancer	2	480

This time in groups, post to #lecture tagging members:

- ~~• Does the model meet anti-classification fairness w.r.t. gender?~~
- Does the model meet group fairness?

$$P[Y' = 1 \mid A = a] = P[Y' = 1 \mid A = b]$$

Equalized Odds

Equalized odds

- Anti-classification (fairness through blindness)
- Group fairness (independence)
- *Equalized odds (separation)*
- ...and numerous others and variations!

Equalized odds

Key idea: Focus on accuracy (not outcomes) across two groups

- Similar default rates on accepted loans across racial/gender groups?
- Similar rate of "bad hires" and "missed stars" between gender groups?
- Similar accuracy of predicted recidivism vs actual recidivism across racial groups?

Accuracy matters, not outcomes!

Equalized odds

$$P[Y'=1 \mid Y=0, A=a] = P[Y'=1 \mid Y=0, A=b]$$

$$P[Y'=0 \mid Y=1, A=a] = P[Y'=0 \mid Y=1, A=b]$$

Statistical property of *separation*: $Y' \perp A \mid Y$

- Prediction must be independent of the sensitive attribute *conditional* on the target variable

Review: Confusion Matrix

		Actual value	
		$Y = 1$	$Y = 0$
Predicted value	$Y' = 1$	True Positive Rate $P[Y' = 1 \mid Y = 1]$	False Positive Rate $P[Y' = 1 \mid Y = 0]$
	$Y' = 0$	False Negative Rate $P[Y' = 0 \mid Y = 1]$	True Negative Rate $P[Y' = 0 \mid Y = 0]$

Can we explain separation in terms of model errors?

- $P[Y'=1 \mid Y=0, A=a] = P[Y'=1 \mid Y=0, A=b]$
- $P[Y'=0 \mid Y=1, A=a] = P[Y'=0 \mid Y=1, A=b]$

Separation

$$P[Y'=1 \mid Y=0, A=a] = P[Y'=1 \mid Y=0, A=b] \text{ (FPR parity)}$$

$$P[Y'=0 \mid Y=1, A=a] = P[Y'=0 \mid Y=1, A=b] \text{ (FNR parity)}$$

- $Y' \perp A \mid Y$: Prediction must be independent of the sensitive attribute *conditional* on the target variable
- i.e., All groups are susceptible to the same false positive/negative rates
- Example: Y' : Promotion decision, A : Gender of applicant: Y : Actual job performance

Testing Separation

Requires realistic representative test data (telemetry or representative test data, not random)

Separately measure false positive and false negative rates

- e.g, for FNR, compare $P[\text{promoted} = 0 \mid \text{female, good employee}]$ vs $P[\text{promoted} = 0 \mid \text{male, good employee}]$

How is this different from testing group fairness?

Breakout Cont': Cancer Prognosis

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In groups, post to #1ecture tagging members:

- ~~Does the model meet anti-classification fairness w.r.t. gender?~~
- ~~Does the model meet group fairness?~~
- Does the model meet equalized odds?
- Is the model fair enough to use?

Metric #1,284.

Okay, the True Positives divided by the False Positives, multiplied by the total number of Negative Predictions, plus the temperature of the room, multiplied by the negative exponential of the number of words in this sentence, should be the same for all sensitive groups.

What are we measuring again?

Fairness.

Right.



Many measures

Many measures proposed, most variations on the three discussed

- Different cells of the confusion matrix (e.g., false discovery rate, lift)

Some consider downstream utility of various outcomes.

Foundation models used for open-ended generation require new evaluation approaches.

- Fairness measured through bias benchmarks (BBQ, StereoSet) and holistic evaluation (HELM)

... Except fairness is impossible

Inherent Trade-Offs in the Fair Determination of Risk Scores

Jon Kleinberg *

Sendhil Mullainathan †

Manish Raghavan ‡

Abstract

Recent discussion in the public sphere about algorithmic classification has involved tension between competing notions of what it means for a probabilistic classification to be fair to different groups. We formalize three fairness conditions that lie at the heart of these debates, and we prove that except in highly constrained special cases, there is no method that can satisfy these three conditions simultaneously. Moreover, even satisfying all three conditions approximately requires that the data lie in an approximate version of one of the constrained special cases identified by our theorem. These results suggest some of the ways in which key notions of fairness are incompatible with each other, and hence provide a framework for thinking about the trade-offs between them.

1 Introduction

There are many settings in which a sequence of people comes before a decision-maker, who must make a judgment about each based on some observable set of features. Across a range of applications, these judgments are being carried out by an increasingly wide spectrum of approaches ranging from human expertise to algorithmic and statistical frameworks, as well as various combinations of these approaches.

Along with these developments, a growing line of work has asked how we should reason about issues of bias and discrimination in settings where these algorithmic and statistical techniques, trained on large datasets of past instances, play a significant role in the outcome. Let us consider three examples where such issues arise, both to illustrate the range of relevant contexts, and to surface some of the challenges.

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(quick pause for illustration)

Recidivism Revisited



[ProPublica article](#)

COMPAS system, developed by Northpointe: Used by judges in sentencing decisions across multiple states (incl. PA)

Which fairness definition?

Table 11.1: COMPAS Fairness Metrics

Metric	Caucasian	African American
False Positive Rate (FPR)	23%	45%
False Negative Rate (FNR)	48%	28%
False Discovery Rate (FDR)	41%	37%

- ProPublica: COMPAS violates equalized odds w/ FPR & FNR
- Northpointe: COMPAS is fair because it has similar FDRs
 - $FDR = FP / (FP + TP) = 1 - \text{Precision}$; $FPR = FP / (FP + TN)$
- **Q. Which measure is appropriate in this context?**

The key insight

Multiple fairness criteria cannot be simultaneously satisfied.

Fairness is fundamentally a value-laden engineering decision requiring stakeholder deliberation, not a technical optimization problem with a single correct solution.

Choosing a Fairness Measure

From model metrics to product decisions

Different fairness metrics reflect different fairness goals, different user groups that we care about, and different trade-offs.

- **Equalized odds – avoid disparate treatment.**
 - "We should just focus on making the right decisions; We are fair by treating everybody equally and making sure everyone gets the opportunity they actually deserve." (equality)
- **Group fairness – avoid disparate impact.**
 - "We should carefully massage our decisions (and maybe be more lenient with giving one group more opportunity) to make sure that we give all the groups the same final outcomes." (equity)

Identifying Fairness Goals is a Requirements Engineering Problem

- What is the goal of the system? What benefits does it provide and to whom?
- Who are the stakeholders of the system? What are the stakeholders' views or expectations on fairness and where do they conflict? Are we trying to achieve fairness based on equality or equity?
- What subpopulations (including minority groups) may be using or be affected by the system? What types of harms can the system cause with discrimination?
- Does fairness undermine any other goals of the system (e.g., accuracy, profits, time to release)?
- Are there legal anti-discrimination requirements to consider? Are there societal expectations about ethics w.r.t. to this product? What is the activist position?

...

Negotiate Fairness Goals/Measures

Negotiate with stakeholders to determine fairness requirement for the product: What is the suitable notion of fairness for the product? Equality or equity?

Map the requirements to model-level (model) specifications: Anti-classification? Group fairness? Equalized odds?

Negotiation can be challenging!

- Conflicts with other system goals (accuracy, profits...)
- Conflicts among different beliefs, values, political views, etc.,

Intuitive Justice: Research on what people perceive as fair (psychology)

- When rewards depend on inputs and participants can choose contributions: Most people find it fair to split rewards proportional to inputs
- Most people agree that for a decision to be fair, personal characteristics that do not influence the reward, such as gender or age, should not be considered when dividing the rewards.

Complexity: Individual and group differences not always clearly attributable, unequal starting positions

Dealing with unequal starting positions

Equality (minimize disparate treatment):

- Treat everybody equally, regardless of starting position
- Focus on meritocracy, strive for fair opportunities
- Equalized-odds-style fairness; equality of opportunity

Equity (minimize disparate impact):

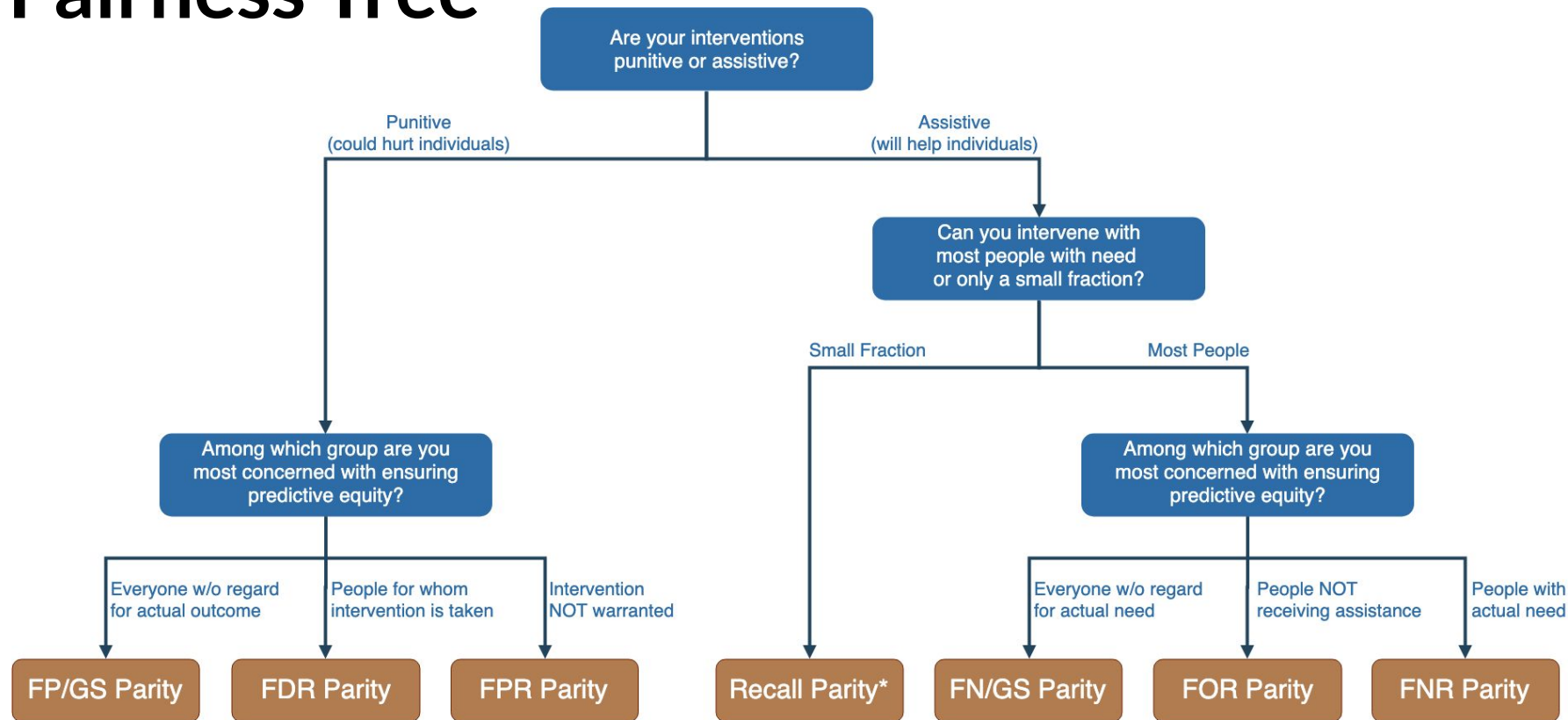
- Lift disadvantaged group, affirmative action
- Strive for similar outcomes (distributive justice)
- Group-fairness-style fairness; equality of outcomes

Each rooted in long history of law/philosophy, and typically incompatible. Problem and goal dependent

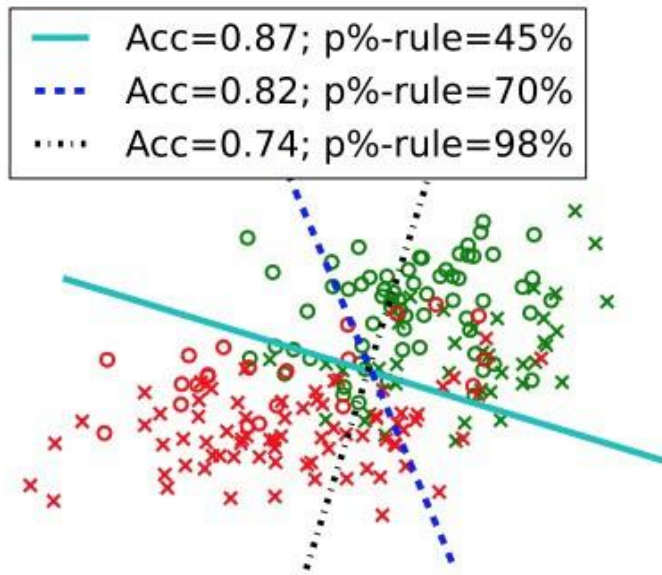
Heuristic: Punitive vs Assistive Decisions

- If the decision is **punitive** in nature:
 - Harm is caused when a group is given an unwarranted penalty
 - e.g. decide whom to deny bail based on risk of recidivism
 - Heuristic: Use a fairness metric (equalized odds) based on **FPR**
- If the decision is **assistive** in nature:
 - Harm is caused when a group in need is denied assistance
 - e.g., decide who should receive a loan or a food subsidy
 - Heuristic: Use a fairness metric based on **FNR**

Fairness Tree



Trade-offs in Fairness vs Accuracy



Fairness imposes constraints, limits what models can be learned

But: Arguably, unfair predictions are not desirable!

Determine how much compromise in accuracy or fairness is acceptable to your stakeholders

[Fairness Constraints: Mechanisms for Fair Classification.](#) Zafar et al. (AISTATS 2017).

Fairness, Accuracy, and Profits

Fairness can conflict with accuracy goals

Fairness can conflict with organizational goals (profits, usability)

Fairer products may attract more customers

Unfair products may receive bad press, reputation damage

Improving fairness through better data can benefit everybody

Recidivism Revisited



[ProPublica article](#)

COMPAS system, developed by Northpointe: Used by judges in sentencing decisions across multiple states (incl. PA)

Reading: people's concerns are much broader than discrimination.

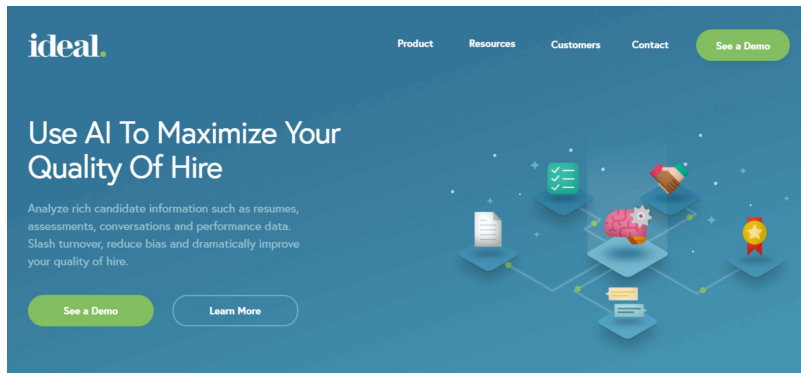
Majority judged half of COMPAS's input features unfair to use.

Finding: People share a moral reasoning process for mapping latent properties to fairness judgments. They *disagree* on the latent properties themselves (does this feature cause the outcome?)

- Latent properties: reliability, relevance, privacy, volitionality, causes outcome, causes vicious cycle, causes disparity in outcomes, and caused by sensitive group membership.

Feature		Mean fairness
1.	Current Charges	6.38
2.	Criminal History: self	6.37
3.	Substance Abuse	4.84
4.	Stability of Employment	4.49
5.	Personality	3.87
6.	Criminal Attitudes	3.63
7.	Neighborhood Safety	3.14
8.	Criminal History: family and friends	2.78
9.	Quality of Social Life & Free Time	2.70
10.	Education & School Behavior	2.70

Breakout: Fairness Goal for Hire Decisions?



Post as a group in #lecture, tagging group members:

- What kind of harm can be caused (to different stakeholders)?
- Fairness goal: Equality or equity? (briefly explain why)
- Metric: Anti-classification, group fairness, or equalized odds (with FPR/FNR)?

Outlook: Building Fair ML-Based Products

Next lecture: Fairness is a *system-wide* concern

- Fairness beyond model predictions (product design, mitigations, data collection)
- Fairness in process and teamwork, barriers and responsibilities
- Documenting fairness at the interface
- Monitoring
- Promoting best practices